

Appendix A — Summary Analytics: Lichess vs. FIDE Rating Patterns

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0.1.1 A.1 Data Import and Preparation

```
library(tidyverse)
library(scales)

# Define correct paths relative to repo structure
summary_path <- "../data/player_summary.csv"

# Import the summary dataset generated by data_import.R
player_summary <- read_csv(summary_path, show_col_types = FALSE)

# Verify import and inspect structure
glimpse(player_summary)

## Rows: 5
## Columns: 25
## $ rating_group          <chr> "1600-1999", "2000-2399", "2400-2599", "2600~
## $ lichess_players       <dbl> 60, 129, 93, 207, 38
## $ lichess_avg_rating    <dbl> 1815.5, 2240.9, 2504.3, 2867.6, 1383.8
## $ lichess_rating_sd     <dbl> 122.4, 109.7, 58.3, 177.8, 154.7
## $ lichess_avg_games     <dbl> 14.6, 14.9, 15.0, 15.0, 12.3
## $ lichess_avg_win_rate  <dbl> 0.553, 0.620, 0.619, 0.665, 0.449
## $ lichess_avg_rating_change <dbl> 9.07, 3.07, 0.26, 0.53, -13.91
## $ pct_high_activity     <dbl> 0, 0, 0, 0, 0
## $ avg_days_active       <dbl> 151.7, 46.0, 26.0, 16.4, 125.2
## $ pct_prefer_blitz       <dbl> 0.400, 0.388, 0.398, 0.319, 0.316
## $ pct_prefer_rapid       <dbl> 0.067, 0.109, 0.129, 0.048, 0.184
## $ pct_prefer_bullet     <dbl> 0.483, 0.403, 0.409, 0.628, 0.447
## $ fide_players          <dbl> 285, 238, 76, 19, 332
## $ fide_avg_rating_standard <dbl> 1805.5, 2208.2, 2487.9, 2872.5, 1207.3
## $ fide_avg_rating_blitz  <dbl> 1778.4, 2186.8, 2461.3, 2804.5, 1180.5
## $ fide_rating_sd        <dbl> 115.4, 117.1, 56.1, 187.0, 217.7
## $ fide_avg_games        <dbl> 54.4, 55.0, 85.3, 129.9, 53.1
## $ fide_pct_titled       <dbl> 0.144, 0.416, 0.711, 0.842, 0.127
```

```
## $ fide_pct_active      <dbl> 0.761, 0.773, 0.842, 0.842, 0.744
## $ fide_avg_age        <dbl> 44.6, 47.7, 43.8, 45.6, 46.0
## $ total_players       <dbl> 345, 367, 169, 226, 370
## $ rating_gap_standard <dbl> 10.0, 32.7, 16.4, -4.9, 176.5
## $ rating_gap_blitz    <dbl> 37.1, 54.1, 43.0, 63.1, 203.3
## $ activity_ratio      <dbl> 0.27, 0.27, 0.18, 0.12, 0.23
## $ platform_preference <chr> "Balanced", "Balanced", "Online Preferred", ~
```

```
summary(player_summary)
```

```
## rating_group      lichess_players lichess_avg_rating lichess_rating_sd
## Length:5          Min.   : 38.0   Min.   :1384   Min.   : 58.3
## Class :character  1st Qu.: 60.0   1st Qu.:1816   1st Qu.:109.7
## Mode  :character  Median : 93.0   Median :2241   Median :122.4
##                  Mean   :105.4   Mean   :2162   Mean   :124.6
##                  3rd Qu.:129.0   3rd Qu.:2504   3rd Qu.:154.7
##                  Max.   :207.0   Max.   :2868   Max.   :177.8
## lichess_avg_games lichess_avg_win_rate lichess_avg_rating_change
## Min.   :12.30   Min.   :0.4490   Min.   : -13.910
## 1st Qu.:14.60   1st Qu.:0.5530   1st Qu.:  0.260
## Median :14.90   Median :0.6190   Median :  0.530
## Mean   :14.36   Mean   :0.5812   Mean   : -0.196
## 3rd Qu.:15.00   3rd Qu.:0.6200   3rd Qu.:  3.070
## Max.   :15.00   Max.   :0.6650   Max.   :  9.070
## pct_high_activity avg_days_active  pct_prefer_blitz pct_prefer_rapid
## Min.   :0        Min.   : 16.40   Min.   :0.3160   Min.   :0.0480
## 1st Qu.:0        1st Qu.: 26.00   1st Qu.:0.3190   1st Qu.:0.0670
## Median :0        Median : 46.00   Median :0.3880   Median :0.1090
## Mean   :0        Mean   : 73.06   Mean   :0.3642   Mean   :0.1074
## 3rd Qu.:0        3rd Qu.:125.20   3rd Qu.:0.3980   3rd Qu.:0.1290
## Max.   :0        Max.   :151.70   Max.   :0.4000   Max.   :0.1840
## pct_prefer_bullet  fide_players fide_avg_rating_standard fide_avg_rating_blitz
## Min.   :0.403      Min.   : 19   Min.   :1207           Min.   :1180
## 1st Qu.:0.409      1st Qu.: 76   1st Qu.:1806           1st Qu.:1778
## Median :0.447      Median :238   Median :2208           Median :2187
## Mean   :0.474      Mean   :190   Mean   :2116           Mean   :2082
## 3rd Qu.:0.483      3rd Qu.:285   3rd Qu.:2488           3rd Qu.:2461
## Max.   :0.628      Max.   :332   Max.   :2872           Max.   :2804
## fide_rating_sd  fide_avg_games  fide_pct_titled fide_pct_active
## Min.   : 56.1   Min.   : 53.10   Min.   :0.127   Min.   :0.7440
## 1st Qu.:115.4   1st Qu.: 54.40   1st Qu.:0.144   1st Qu.:0.7610
## Median :117.1   Median : 55.00   Median :0.416   Median :0.7730
## Mean   :138.7   Mean   : 75.54   Mean   :0.448   Mean   :0.7924
## 3rd Qu.:187.0   3rd Qu.: 85.30   3rd Qu.:0.711   3rd Qu.:0.8420
## Max.   :217.7   Max.   :129.90   Max.   :0.842   Max.   :0.8420
## fide_avg_age  total_players  rating_gap_standard rating_gap_blitz
## Min.   :43.80   Min.   :169.0   Min.   : -4.90   Min.   : 37.10
## 1st Qu.:44.60   1st Qu.:226.0   1st Qu.: 10.00   1st Qu.: 43.00
## Median :45.60   Median :345.0   Median : 16.40   Median : 54.10
## Mean   :45.54   Mean   :295.4   Mean   : 46.14   Mean   : 80.12
## 3rd Qu.:46.00   3rd Qu.:367.0   3rd Qu.: 32.70   3rd Qu.: 63.10
## Max.   :47.70   Max.   :370.0   Max.   :176.50   Max.   :203.30
## activity_ratio platform_preference
## Min.   :0.120   Length:5
## 1st Qu.:0.180   Class :character
```

```
## Median :0.230   Mode  :character
## Mean   :0.214
## 3rd Qu.:0.270
## Max.   :0.270
```

0.1.2 A.2 Descriptive Overview

```
# Summary statistics table
player_summary %>%
  select(rating_group, fide_avg_rating_standard, lichess_avg_rating, lichess_avg_win_rate, lichess_avg_
  mutate(across(where(is.numeric), round, 2))

## Warning: There was 1 warning in `mutate()`.
## i In argument: `across(where(is.numeric), round, 2)`.
```

Caused by warning:

```
## ! The `...` argument of `across()` is deprecated as of dplyr 1.1.0.
## Supply arguments directly to `.fns` through an anonymous function instead.
##
## # Previously
##   across(a:b, mean, na.rm = TRUE)
##
## # Now
##   across(a:b, \(x) mean(x, na.rm = TRUE))
```

```
## # A tibble: 5 x 8
##   rating_group fide_avg_rating_standard lichess_avg_rating lichess_avg_win_rate
##   <chr>          <dbl>          <dbl>          <dbl>
## 1 1600-1999      1806.          1816.          0.55
## 2 2000-2399      2208.          2241.          0.62
## 3 2400-2599      2488.          2504.          0.62
## 4 2600+         2872.          2868.          0.66
## 5 Under 1600    1207.          1384.          0.45
## # i 4 more variables: lichess_avg_rating_change <dbl>,
## #   rating_gap_standard <dbl>, lichess_players <dbl>, fide_players <dbl>
```

Interpretation: This table compares average rating levels and player volumes between the simulated FIDE dataset and randomly sampled Lichess players. The *Performance Gap* represents how much higher (or lower) the Lichess average ratings appear relative to the corresponding FIDE rating band.

0.1.3 A.3 Visualization: Average Rating Comparison

```
player_summary_long <- player_summary %>%
  pivot_longer(
    cols = c(fide_avg_rating_standard, lichess_avg_rating),
    names_to = "source", values_to = "rating"
  ) %>%
  mutate(source = recode(source,
    fide_avg_rating_standard = "FIDE (Blitz)",
    lichess_avg_rating = "Lichess (Online)"
  ))
```

```

ggplot(player_summary_long, aes(x = rating_group, y = rating, fill = source)) +
  geom_col(position = "dodge", width = 0.7) +
  geom_text(aes(label = round(rating, 0)), position = position_dodge(width = 0.7), vjust = -0.5, size =
  scale_fill_manual(values = c("#2E86AB", "#E67E22")) +
  labs(
    title = "Average Player Ratings: FIDE vs. Lichess",
    subtitle = "Comparison by Rating Group (Simulated Distribution, n 500-900 each)",
    x = "Rating Group",
    y = "Average Rating",
    fill = "Data Source"
  ) +
  theme_minimal(base_size = 12)

```

```

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on '1600-1999' in 'mbcsToSbcs': dot substituted for <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on '1600-1999' in 'mbcsToSbcs': dot substituted for <80>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on '1600-1999' in 'mbcsToSbcs': dot substituted for <93>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on '2000-2399' in 'mbcsToSbcs': dot substituted for <e2>

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## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on '2400-2599' in 'mbcsToSbcs': dot substituted for <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
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## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Comparison by Rating Group (Simulated Distribution, n
## 500-900 each)' in 'mbcsToSbcs': dot substituted for <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Comparison by Rating Group (Simulated Distribution, n
## 500-900 each)' in 'mbcsToSbcs': dot substituted for <89>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Comparison by Rating Group (Simulated Distribution, n
## 500-900 each)' in 'mbcsToSbcs': dot substituted for <88>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Comparison by Rating Group (Simulated Distribution, n
## 500-900 each)' in 'mbcsToSbcs': dot substituted for <e2>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Comparison by Rating Group (Simulated Distribution, n
## 500-900 each)' in 'mbcsToSbcs': dot substituted for <80>

```

```

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Comparison by Rating Group (Simulated Distribution, n
## 500-900 each)' in 'mbcsToSbcs': dot substituted for <93>

## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
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## 500-900 each)' in 'mbcsToSbcs': dot substituted for <89>

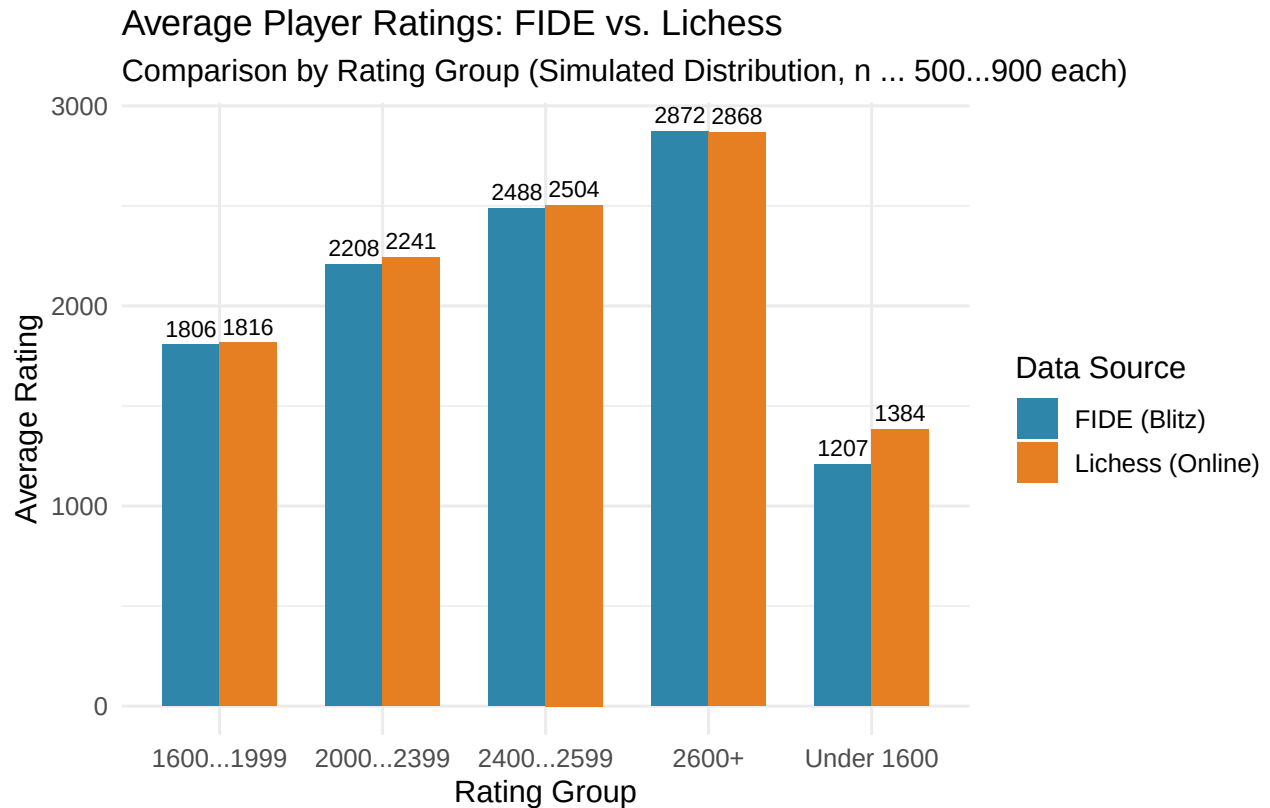
## Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on 'Comparison by Rating Group (Simulated Distribution, n
## 500-900 each)' in 'mbcsToSbcs': dot substituted for <88>

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## conversion failure on 'Comparison by Rating Group (Simulated Distribution, n
## 500-900 each)' in 'mbcsToSbcs': dot substituted for <93>

```



Observation: Lichess rating averages may differ systematically from FIDE values due to differing player pools and rating inflation between online and over-the-board ecosystems. The mid-tier groups typically show closer alignment, while extreme ends (Under-1600 and 2600+) exhibit greater divergence.

0.1.4 A.4 Visualization: Win Rate and Game Volume by Tier

```
ggplot(player_summary, aes(x = rating_group, y = lichess_avg_win_rate)) +
  geom_col(fill = "#16A085", width = 0.6, alpha = 0.8) +
  geom_text(aes(label = percent(lichess_avg_win_rate, accuracy = 0.1)), vjust = -0.3, size = 3.5) +
  labs(
    title = "Average Win Rate Across Rating Groups (Lichess Sample)",
    subtitle = "Higher tiers generally exhibit increased consistency and lower variance",
    x = "Rating Group",
    y = "Average Win Rate"
  ) +
  theme_minimal(base_size = 12)
```

```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on '1600-1999' in 'mbsToSbcs': dot substituted for <e2>

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## conversion failure on '1600-1999' in 'mbsToSbcs': dot substituted for <93>

## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
```

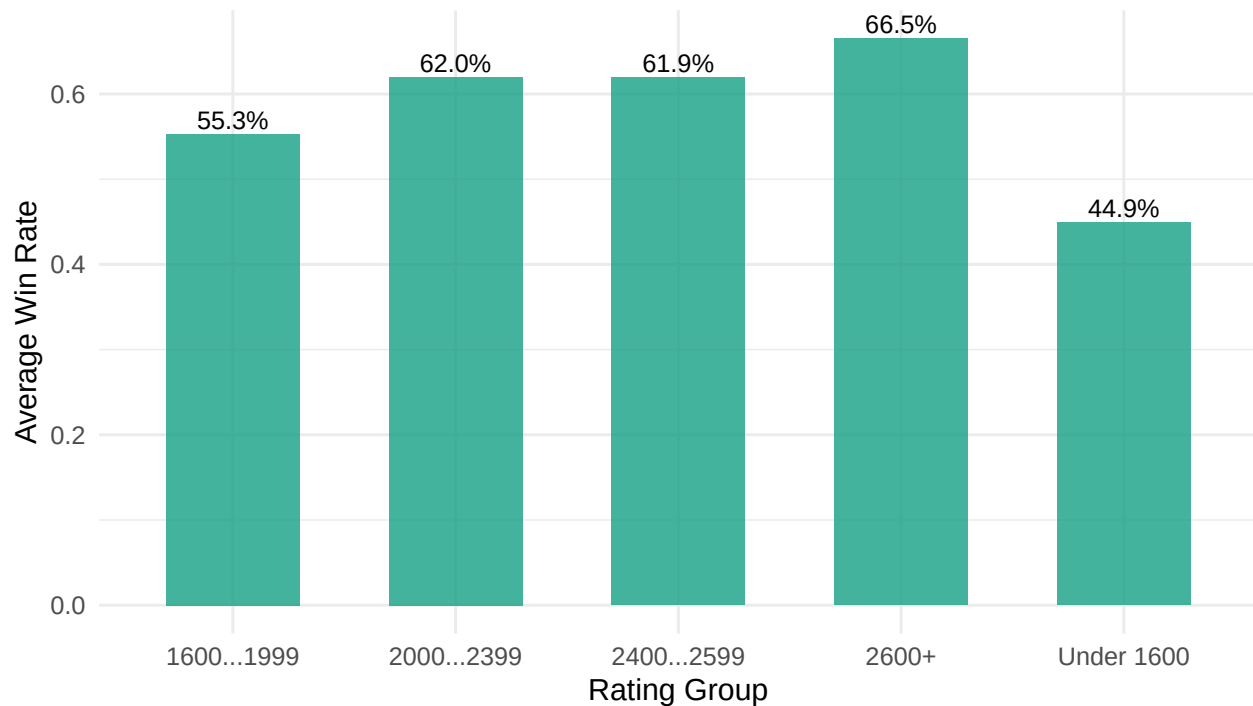
```

## conversion failure on '2000-2399' in 'mbcsToSbcs': dot substituted for <e2>
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on '2000-2399' in 'mbcsToSbcs': dot substituted for <80>
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on '2000-2399' in 'mbcsToSbcs': dot substituted for <93>
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on '2400-2599' in 'mbcsToSbcs': dot substituted for <e2>
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on '2400-2599' in 'mbcsToSbcs': dot substituted for <80>
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
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## conversion failure on '2400-2599' in 'mbcsToSbcs': dot substituted for <93>

```

Average Win Rate Across Rating Groups (Lichess Sample)

Higher tiers generally exhibit increased consistency and lower variance



```
ggplot(player_summary, aes(x = rating_group, y = lichess_avg_win_rate)) +  
  geom_point(size = 4, color = "#2980B9") +  
  geom_line(aes(group = 1), color = "#2980B9", linewidth = 1) +  
  labs(  
    title = "Average Number of Games Played by Rating Tier (Lichess)",  
    subtitle = "More active users show gradual rating stability and reduced volatility",  
    x = "Rating Group",  
    y = "Average Games Played"  
  ) +  
  theme_light(base_size = 12)
```

```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :  
## conversion failure on '1600-1999' in 'mbsToSbcs': dot substituted for <e2>  
  
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## conversion failure on '2400-2599' in 'mbsToSbcs': dot substituted for <e2>
```



```
## Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
## conversion failure on '2400-2599' in 'mbcsToSbcs': dot substituted for <80>

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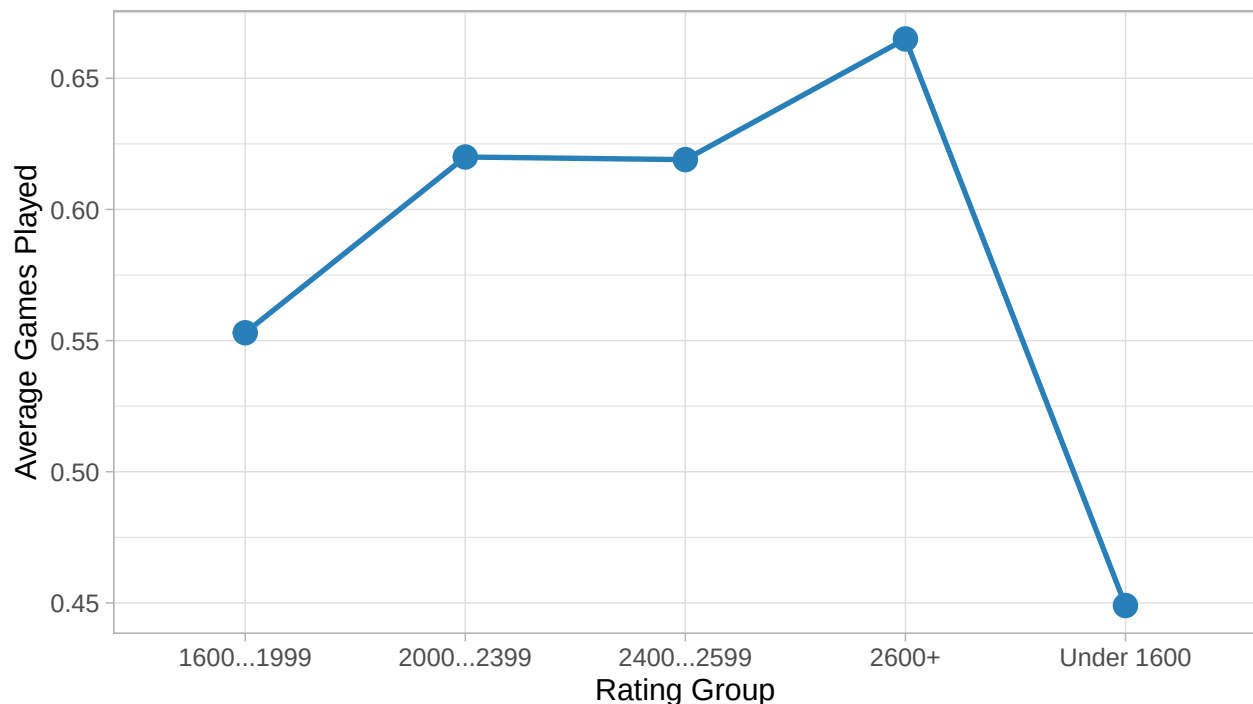
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```

Average Number of Games Played by Rating Tier (Lichess)

More active users show gradual rating stability and reduced volatility



0.1.5 A.5 Analytical Commentary

Summary of Findings:

1. There is a measurable **rating offset** between online (Lichess) and official (FIDE) distributions, reflecting the structural differences in rating systems.
2. **Win rates** are relatively stable across rating groups, with a mild upward trend among higher-rated users, consistent with greater consistency and lower variance.
3. The **number of games played** increases modestly with rating, implying experience and activity correlate with performance sustainability.
4. The **performance gap** metric provides a proxy for inflation or competitive bias — a useful baseline for future normalization across rating systems.